# Ruei-Che Chang

2474 Stone Rd
Ann Arbor, MI, USA
✉ rueiche@umich.edu
rueichechang.github.io

## Education

2022–present **University of Michigan**, *Ann Arbor, Michigan*.
**Ph.D. Student in Computer Science & Engineering**.
*Human-AI Lab, advised by Anhong Guo.*

2020–2022 **National Taiwan University**, *Taipei, Taiwan*.
**Visiting Student and Research Assistant**.
*Interactive Graphics Lab, advised by Robin Bing-Yu Chen*.

2019–2021 **Dartmouth College**, *New Hampshire, Hanover*.
**M.S. in Computer Science**.
Thesis: Adaptive Add-on Circuit Components for People who are Blind or Have Low Vision.
*XDiscovery lab, advised by Xing-Dong Yang*.

2014–2018 **National Cheng Kung University**, *Tainan Taiwan*.
**B.S. in Electrical Engineering.**

## Publications (* denote equal contribution)

2022 [C.7] **Ruei-Che Chang**, Chao-Hsien Ting, Chia-Sheng Hung, Wan-Chen Lee, Liang-Jin Chen, Yu-Tze Chao, Bing-Yu Chen, Anhong Guo, "OmniScribe: Authoring Immersive Audio Description for 360° Videos." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'22**).

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen, "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'21**).

[C.5] **Ruei-Che Chang***, Wen-Ping Wang*, Chi-Huan Chiang, Te-Yen Wu, Zheer Xu, Justin Luo, Bing-Yu Chen, Xing-Dong Yang, "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'21**).

2020 [C.4] **Ruei-Che Chang***, Chi-Huan Chiang*, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen, "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Proceedings of the ACM Symposium on Spatial User Interaction* (**SUI'20**).

[C.3] **Ruei-Che Chang***, Yi-Shyuan Chiang*, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang, "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**).

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen, "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**) **[Honorable Mention Award, Top 5%]**.

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen, "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'19**).

## Awards

2020 **Honorable Mention Award for Glissade [C.2]** at *ACM CHI 2020.*

2018 Taiwan internship scholarship from the Ministry of Education (21.53% acceptance rate).

2017 2nd Prize in Taiwan Intelligent Robot Contest (30 entries).

## Research Experience

Sep 2021 – present **Human-AI Lab, University of Michigan**
***Remote research intern, advised by Anhong Guo.***

- Developed a web authoring interface to support video describers to understand and describe 360° videos, and a mobile prototype enabling blind people to consume the generated descriptions. [C.7]

Jun 2020 – Jul 2022 **Interactive Graphics Lab, National Taiwan University**
***Research Assistant, advised by Robin Bing-Yu Chen and Tom Yeh.***

- Leading weekly two-hours group discussion on different HCI topics; Mentoring master students on brainstorming and shaping their research projects aimed for top HCI venues.
- Designed and conducted a study to identify accessibility challenges of laser-cut assembly; Implemented Fusion 360 plug-in to generate tactile aids for input laser-cutting models; Evaluated tactile aids, and leading video and qualitative analysis [C.6].
- Designed and conducted a user study with blind and sighted groups to understand (non-)visual strategies of laser-cut assembly; Conducted inter-coding video analysis.

Feb 2019 – Aug 2019 ***Research Assistant, advised by Robin Bing-Yu Chen and Da-Yuan Huang.***
Implemented six tablet applications for Glissade[C.2] using Unity, and provided Masque [C.1] designers with authoring tools to specify haptic feedback on the tablet.

- Integrated arduino software and electronics; Implemented two VR applications; Designed and conducted user research to obtain initial feedback on using TanGo [C.4].

Sep 2019 – Sep 2020 **XDiscovery Lab, Dartmouth College**
***Graduate student, advised by Xing-Dong Yang.***

- Designed and fabricated haptically-assisted conductive 3D-printed adaptations (via Fusion 360) for existing circuit components that can interact with smartphones via a conductive breadboard extension enclosures; Designed and conducted a study to identify accessibility challenges for circuit prototyping, and another study to evaluate designed adaptations [C.5].
- Collected and labeled 500+ images of breadboard circuits as training data for Mask-RCNN. Computer vision algorithms were implemented to segment the holes and electronics pins on the breadboard, and to provide on-screen AR assistance to students via Android smartphones.

Sep 2018 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
***Research Assistant, advised by Yung-Ju (Stanley) Chang.***

- Designed and executed an experiment using methods: WoZ, speed dating, user enactment and contextual inquiry, to understand human-AI communication in smart home [C.3].

## Work Experience

2018 **Mandatory Military Service**, Taiwan.

2018 **Undergraduate research intern** at *Industrial Technology Research Institute (ITRI)*, Hsinchu, Taiwan.

## Invited Talks/Hosts

**Note: TAICHI is the top domestic HCI conference in Taiwan.**

TAICHI'21 Invited to co-host a panel discussing research culture with 3 international PhD students.

TAICHI'21 Invited to present *Daedalus*[*C.6*] and *AccessibleCircuits*[*C.5*] at the Best of HCI session.

## Reviewer

| | |
|---|---|
| CHI | '22 |
| UIST | '21 '22 |
| DIS | '22 |
| ISS | '22 |
| MobileHCI | '22 |
| LBW at CHI | '20 '21 '22 |

## Student Mentoring

2021–2022 **Fang-Ying Liao**, *Master student* at National Taiwan University.

2021–2022 **Chao-Hsien Ting**, *Master student* at National Taiwan University. (OmniScribe [C.7], UIST'22)

Jan–Apr 2021 **Chih-An Tsao**, *Master student* at National Taiwan University. (Daedalus [C.6], UIST'21)

## Skills

| | |
|---|---|
| Programming | Java, C#, Python, JavaScript, ROS, OpenCV, MongoDB, Swift, LaTeX |
| Prototyping | Arduino, 3D-printing, Fusion 360, Laser-cutting |
| Platforms/IDE | Unity3D, Android Studio, Fusion 360, Xcode |

## References

**Anhong Guo**
*Assistant Professor, Department of Computer Science*
University of Michigan, Ann Arbor
✉ anhong@umich.edu
(*Personal Webpage*)

**Robin Bing-Yu Chen**
*Distinguished Professor, Department of Computer Science & Information Engineering*
National Taiwan University
✉ robin@ntu.edu.tw
(*Personal Webpage*)

**Xing-Dong Yang**
*Associate Professor, Department of Computer Science*
Simon Fraser University
✉ xing-dong.yang@dartmouth.edu
(*Personal Webpage*)

**Tom Yeh**
*Associate Professor, Department of Computer Science*
University of Colorado
✉ tom.yeh@colorado.edu
(*Personal Webpage*)